1 Equations in ρ coordinates

$$0 = \frac{1}{V'}\frac{\partial}{\partial\rho}\left[V'\langle|\nabla\rho|^2\rangle\frac{\partial}{\partial\rho}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_\mathrm{e} + Z_\mathrm{i}en_\mathrm{i} + Z_\mathrm{z}en_\mathrm{z} + Z_\mathrm{b}en_\mathrm{b} + Z_\mathrm{b}egn_\mathrm{b}^\mathrm{rp}\right) \tag{1.1}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi}_t = {} & V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\left[\frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\psi_t\right] \\ & + \frac{\mu_0}{\langle B^\theta\rangle}\left[-en_\mathrm{e}\left(\langle Bu_{\mathrm{e}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle\right) + Z_\mathrm{i}en_\mathrm{i}\left(\langle Bu_{\mathrm{i}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle\right)\right. \\ & \left. + Z_\mathrm{z}en_\mathrm{z}\left(\langle Bu_{\mathrm{z}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right) + Z_\mathrm{b}en_\mathrm{b}\left(\langle Bu_{\mathrm{b}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{1.2}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi} = {} & \frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\left[V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\psi\right] \\ & - \frac{\mu_0}{\langle R^{-2}\rangle}\left(-en_\mathrm{e}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle + Z_\mathrm{i}en_\mathrm{i}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle + Z_\mathrm{z}en_\mathrm{z}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + Z_\mathrm{b}en_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\end{aligned} \tag{1.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{1.4}$$

$$\frac{\partial}{\partial t}\psi_t = \dot{\psi}_t \tag{1.5}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V'n_\mathrm{e}) = {} & -\frac{1}{V'}\frac{\partial}{\partial\rho}[V'n_\mathrm{e}(u_\mathrm{e}^\rho - u_g^\rho)] + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\ & + Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e}-n_\mathrm{esup}) + (1-f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{1.6}$$

$$\begin{aligned}0 = {} & -(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}n_\mathrm{e}T_\mathrm{e} + en_\mathrm{e}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}\Phi \\ & - e\frac{\partial\psi}{\partial\rho}In_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + e\frac{\partial\psi}{\partial\rho}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle\end{aligned} \tag{1.7}$$

$$\begin{aligned}m_\mathrm{e}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\parallel}\rangle = {} & -\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}} + \hat{\mu}_2^\mathrm{e}BV_{2\mathrm{e}} \\ & + \ell_{11}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\parallel}\rangle \\ & - \ell_{12}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\ & - en_\mathrm{e}\langle BE_\parallel\rangle - \nu_\mathrm{L}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \nu_{0\mathrm{e}}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle \\ & - \frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\\
&-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle-\frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\\
&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right] \qquad (1.8)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'm_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle)=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle+(u_{\mathrm{e}}^{\rho}-u_g^{\rho})m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\langle Ru_{\mathrm{e}\zeta}\rangle+\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\right]\\
&+\ell_{11}^{\mathrm{ee}}\langle Ru_{\mathrm{e}\zeta}\rangle+\ell_{11}^{\mathrm{ei}}\langle Ru_{\mathrm{i}\zeta}\rangle+\ell_{11}^{\mathrm{ez}}\langle Ru_{\mathrm{z}\zeta}\rangle+\ell_{11}^{\mathrm{eb}}\langle Ru_{\mathrm{b}\zeta}\rangle\\
&-\ell_{12}^{\mathrm{ee}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle-\ell_{12}^{\mathrm{ei}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle-\ell_{12}^{\mathrm{ez}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle\\
&-en_{\mathrm{e}}\langle RE_{\zeta}\rangle-e\frac{\partial\psi}{\partial\rho}n_{\mathrm{e}}u_{\mathrm{e}}^{\rho}\\
&-\frac{e\langle|\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}}\frac{\partial\psi}{\partial\rho}\left[(1-C_{\mathrm{T}})T_{\mathrm{e}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}+C_{\mathrm{T}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}+n_{\mathrm{e}}T_{\mathrm{e}}\frac{\partial\psi}{\partial\rho}C_{\mathrm{p}}\right]\\
&-\nu_{\mathrm{L}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle-\nu_{0\mathrm{e}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle \qquad (1.9)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{1}{V'^{5/3}}\frac{\partial}{\partial t}(V'^{5/3}n_{\mathrm{e}}T_{\mathrm{e}})=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[-\chi_{\mathrm{e}}\langle|\nabla\rho|^2\rangle n_{\mathrm{e}}\frac{\partial}{\partial\rho}T_{\mathrm{e}}+V_{\mathrm{e}}^{\mathrm{hp}}\langle|\nabla\rho|\rangle n_{\mathrm{e}}T_{\mathrm{e}}+\frac{5}{2}n_{\mathrm{e}}T_{\mathrm{e}}(u_{\mathrm{e}}^{\rho}-u_g^{\rho})\right]\\
&-u_g^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}-u_{\mathrm{i}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{i}}T_{\mathrm{i}}-u_{\mathrm{z}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{z}}T_{\mathrm{z}}-\hat{u}_{\mathrm{i}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{i}}\rangle-\hat{u}_{\mathrm{z}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle\\
&-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{i}}}{\tau_{\mathrm{ei}}}-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{z}}}{\tau_{\mathrm{ez}}}\\
&-\dot{\psi}_t\langle B^{\theta}\rangle(-en_{\mathrm{e}}\hat{u}_{\mathrm{e}\theta}+Z_{\mathrm{i}}en_{\mathrm{i}}\hat{u}_{\mathrm{i}\theta}+Z_{\mathrm{z}}en_{\mathrm{z}}\hat{u}_{\mathrm{z}\theta})\\
&+\dot{\psi}\left(-en_{\mathrm{e}}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle+Z_{\mathrm{i}}en_{\mathrm{i}}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle+Z_{\mathrm{z}}en_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right)\\
&-\nu_{\mathrm{L}}T_{\mathrm{e}}(n_{\mathrm{e}}-n_{\mathrm{esup}})-\frac{3}{2}\nu_{\mathrm{L}T_{\mathrm{e}}}n_{\mathrm{e}}(T_{\mathrm{e}}-T_{\mathrm{esup}})\\
&-E_{\mathrm{H}}\nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\
&+f_{\mathrm{ei}}P_{\mathrm{b}}+P_{\alpha\mathrm{e}}+P_{\mathrm{RFe}}-P_{\mathrm{Br}} \qquad (1.10)
\end{aligned}
$$

$$
\frac{5}{2}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle=-\hat{\mu}_2^{\mathrm{e}}\langle Bu_{\mathrm{e}\parallel}\rangle-\hat{\mu}_3^{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle+\hat{\mu}_2^{\mathrm{e}}BV_{1\mathrm{e}}+\hat{\mu}_3^{\mathrm{e}}BV_{2\mathrm{e}}
$$

$$
\begin{aligned}
&- \ell_{21}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\|}\rangle - \ell_{21}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\|}\rangle - \ell_{21}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\|}\rangle - \ell_{21}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\|}\rangle \\
&+ \ell_{22}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \ell_{22}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \ell_{22}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle
\end{aligned} \tag{1.11}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{i}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{i}}(u_{\mathrm{i}}^{\rho} - u_g^{\rho})] + \nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\
&- \nu_{\mathrm{L}}(n_{\mathrm{i}} - n_{\mathrm{isup}}) + \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}}\frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
0 = &-(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} - Z_{\mathrm{i}} e n_{\mathrm{i}}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho} I\langle Bu_{\mathrm{i}\|}\rangle - Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho}\langle B^2\rangle\langle Ru_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.13}
$$

$$
\begin{aligned}
m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial t}\langle Bu_{\mathrm{i}\|}\rangle = &-\hat{\mu}_1^{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \hat{\mu}_2^{\mathrm{i}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\mu}_1^{\mathrm{i}} BV_{1\mathrm{i}} + \hat{\mu}_2^{\mathrm{i}} BV_{2\mathrm{i}} \\
&+ \ell_{11}^{\mathrm{ie}}\langle Bu_{\mathrm{e}\|}\rangle + \ell_{11}^{\mathrm{ii}}\langle Bu_{\mathrm{i}\|}\rangle + \ell_{11}^{\mathrm{iz}}\langle Bu_{\mathrm{z}\|}\rangle + \ell_{11}^{\mathrm{ib}}\langle Bu_{\mathrm{b}\|}\rangle \\
&- \ell_{12}^{\mathrm{ie}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \ell_{12}^{\mathrm{ii}}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \ell_{12}^{\mathrm{iz}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle BE_{\|}\rangle - \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&+ \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle - \frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right]
\end{aligned} \tag{1.14}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle + (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\langle Ru_{\mathrm{i}\zeta}\rangle + \langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle\right] \\
&+ \ell_{11}^{\mathrm{ie}}\langle Ru_{\mathrm{e}\zeta}\rangle + \ell_{11}^{\mathrm{ii}}\langle Ru_{\mathrm{i}\zeta}\rangle + \ell_{11}^{\mathrm{iz}}\langle Ru_{\mathrm{z}\zeta}\rangle + \ell_{11}^{\mathrm{ib}}\langle Ru_{\mathrm{b}\zeta}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&- \ell_{12}^{\mathrm{ie}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle - \ell_{12}^{\mathrm{ii}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle - \ell_{12}^{\mathrm{iz}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle RE_{\zeta}\rangle + Z_{\mathrm{i}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{i}} u_{\mathrm{i}}^{\rho} \\
&+ Z_{\mathrm{i}} \frac{e\langle |\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left[(1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle - \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle \\
&- \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.15}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{i}} T_{\mathrm{i}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[-\chi_{\mathrm{i}} \langle |\nabla\rho|^2\rangle n_{\mathrm{i}} \frac{\partial}{\partial \rho} T_{\mathrm{i}} + V_{\mathrm{i}}^{\mathrm{hp}} \langle |\nabla\rho|\rangle n_{\mathrm{i}} T_{\mathrm{i}} + \frac{5}{2} n_{\mathrm{i}} T_{\mathrm{i}} (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{e}}}{\tau_{\mathrm{ie}}} - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{z}}}{\tau_{\mathrm{iz}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{i}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{i}}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \langle B u_{\mathrm{b}\parallel}\rangle \frac{\langle B u_{\mathrm{i}\parallel}\rangle}{\langle B^2\rangle} + P_{\alpha\mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{1.16}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{i}} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle = &- \hat{\mu}_2^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel}\rangle - \hat{\mu}_3^{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \hat{\mu}_2^{\mathrm{i}} B V_{1\mathrm{i}} + \hat{\mu}_3^{\mathrm{i}} B V_{2\mathrm{i}} \\
&- \ell_{21}^{\mathrm{ie}} \langle B u_{\mathrm{e}\parallel}\rangle - \ell_{21}^{\mathrm{ii}} \langle B u_{\mathrm{i}\parallel}\rangle - \ell_{21}^{\mathrm{iz}} \langle B u_{\mathrm{z}\parallel}\rangle - \ell_{21}^{\mathrm{ib}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&+ \ell_{22}^{\mathrm{ie}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \ell_{22}^{\mathrm{ii}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \ell_{22}^{\mathrm{iz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle
\end{aligned} \tag{1.17}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho})] + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\
&- \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}}
\end{aligned} \tag{1.18}
$$

$$
\begin{aligned}
0 = &- (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} - Z_{\mathrm{z}} e n_{\mathrm{z}} (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} \Phi \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{z}\parallel}\rangle - Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} \langle B^2\rangle \langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{1.19}
$$

$$
\begin{aligned}
m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel}\rangle = &- \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel}\rangle - \hat{\mu}_2^{\mathrm{z}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle + \hat{\mu}_1^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_2^{\mathrm{z}} B V_{2\mathrm{z}} \\
&+ \ell_{11}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{zi}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{zz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle B E_{\parallel} \rangle - \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{\langle B^2 \rangle}{I} \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right] \qquad (1.20)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \langle R u_{\mathrm{z}\zeta} \rangle + \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \right] \\
&+ \ell_{11}^{\mathrm{ze}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{zi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{zz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{zb}} \langle R u_{\mathrm{b}\zeta} \rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle R \hat{q}_{\mathrm{e}\zeta} \rangle - \ell_{12}^{\mathrm{zi}} \langle R \hat{q}_{\mathrm{i}\zeta} \rangle - \ell_{12}^{\mathrm{zz}} \langle R \hat{q}_{\mathrm{z}\zeta} \rangle \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle R E_{\zeta} \rangle + Z_{\mathrm{z}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{z}} u_{\mathrm{z}}^{\rho} \\
&+ Z_{\mathrm{z}} \frac{e \langle |\nabla \rho|^2 \rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} \left[(1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \qquad (1.21)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{z}} T_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[- \chi_{\mathrm{z}} \langle |\nabla \rho|^2 \rangle n_{\mathrm{z}} \frac{\partial}{\partial \rho} T_{\mathrm{z}} + V_{\mathrm{z}}^{\mathrm{hp}} \langle |\nabla \rho| \rangle n_{\mathrm{z}} T_{\mathrm{z}} + \frac{5}{2} n_{\mathrm{z}} T_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{e}}}{\tau_{\mathrm{ze}}} - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{i}}}{\tau_{\mathrm{zi}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{z}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{z}}} n_{\mathrm{z}} (T_{\mathrm{z}} - T_{\mathrm{zsup}}) \\
&+ P_{\alpha \mathrm{z}} + P_{\mathrm{RFz}} \qquad (1.22)
\end{aligned}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle = &- \hat{\mu}_2^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\mu}_3^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_3^{\mathrm{z}} B V_{2\mathrm{z}} \\
&- \ell_{21}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel} \rangle - \ell_{21}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel} \rangle - \ell_{21}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel} \rangle - \ell_{21}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \ell_{22}^{\mathrm{ze}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \ell_{22}^{\mathrm{zi}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \ell_{22}^{\mathrm{zz}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle
\end{aligned}
$$

$$- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\|} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\|} \rangle \tag{1.23}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{b}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{b}}(u_{\mathrm{b}}^{\rho} - u_g^{\rho})] + \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} - g\frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} - \nu_{\mathrm{b}} n_{\mathrm{b}} - \nu_{\mathrm{bL}} n_{\mathrm{b}} - \nu_{\mathrm{eff}} g(\sqrt{\delta} n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}}) \\ &+ \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}\end{aligned} \tag{1.24}$$

$$\begin{aligned}0 = &-(\langle B^2 \rangle \langle R^2 \rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{b}} T_{\mathrm{b}} - Z_{\mathrm{b}} e n_{\mathrm{b}} (\langle B^2 \rangle \langle R^2 \rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\ &+ Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{b}\|} \rangle - Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} \langle B^2 \rangle \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned} \tag{1.25}$$

$$\begin{aligned}m_{\mathrm{b}} \frac{\partial}{\partial t} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle = &-(\hat{\mu}_1^{\mathrm{b}} - \ell_{11}^{\mathrm{bb}}) \langle B u_{\mathrm{b}\|} \rangle + \sum_{a \neq \mathrm{b}} \ell_{11}^{\mathrm{b}a} \langle B u_{a\|} \rangle + \hat{\mu}_1^{\mathrm{b}} B V_{1\mathrm{b}} + Z_{\mathrm{b}} e n_{\mathrm{b}} \langle B E_{\|}^{A} \rangle \\ &- \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{0\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + \langle B S_{\mathrm{m}\|\mathrm{b}} \rangle \\ &- \nu_{\mathrm{eff}} g \sqrt{\delta} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle m_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}\end{aligned} \tag{1.26}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V'(u_{\mathrm{b}}^{\rho} - u_g^{\rho}) m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle] \\ &+ \ell_{11}^{\mathrm{be}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{bi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{bz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{bb}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\ &+ Z_{\mathrm{b}} e n_{\mathrm{b}} \dot{\psi} + Z_{\mathrm{b}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{b}} u_{\mathrm{b}}^{\rho} - \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle \\ &- \nu_{0\mathrm{i}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle + \frac{I}{\langle B^2 \rangle} \langle B S_{\mathrm{m}\|\mathrm{b}} \rangle\end{aligned} \tag{1.27}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{01}) = &\frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{01} \frac{\partial}{\partial \rho} n_{01} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{01} \\ &- \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} + \gamma_0 \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_{\mathrm{i}}^{\mathrm{puff}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}}\end{aligned} \tag{1.28}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{02}) = \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{02} \frac{\partial}{\partial \rho} n_{02} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{02} + \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} \tag{1.29}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{03}) = \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{03} \frac{\partial}{\partial \rho} n_{03} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{03} + f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \tag{1.30}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' \bar{n}_z) = &\frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{nz} \frac{\partial}{\partial \rho} \bar{n}_z + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \Delta D_{\mathrm{C0}} \frac{\partial}{\partial \rho} n_{\mathrm{C0}} - n_{\mathrm{e}} \bar{n}_z f_5 S_5 + n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\ &+ \gamma_{0\mathrm{z}} \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + \gamma_{\mathrm{i,z}} \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - (1 - \gamma_{0\mathrm{z}}) \nu_{\mathrm{L}} (\bar{n}_z - n_{\mathrm{C0}}) \\ &+ \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}_{\mathrm{z}}^{\mathrm{puff}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}}\end{aligned} \tag{1.31}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\rm b}^{\rm rp}) = \frac{P_{\rm b\perp}}{E_{\rm b}} + \nu_{\rm eff}(\sqrt{\delta} n_{\rm b} - n_{\rm b}^{\rm rp}) - \nu_{\rm b} n_{\rm b}^{\rm rp} - \frac{1}{V'}\frac{\partial}{\partial \rho} V' \left(\langle |\nabla \rho| \rangle u_{\rm b}^{\rm rp} n_{\rm b}^{\rm rp} - \langle |\nabla \rho|^2 \rangle D_{\rm b}^{\rm rp} \frac{\partial}{\partial \rho} n_{\rm b}^{\rm rp} \right) \quad (1.32)$$

2 Additional important relationship

$$\bar{n}_z = \sum_{i=0}^{\max(Z)-1} n_i \quad (2.1)$$

$$B_p = \frac{|\nabla \rho|}{R}\frac{\partial \psi}{\partial \rho}, \quad \langle B_p^2 \rangle = \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.2)$$

$$B_t = \frac{I}{R} = \frac{1}{R}\frac{4\pi^2}{\langle R^{-2} \rangle}\frac{\partial \psi_t}{\partial V}, \quad \langle B_t^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle}\left(\frac{\partial \psi_t}{\partial V} \right)^2 \quad (2.3)$$

$$\langle B^2 \rangle = \langle B_t^2 \rangle + \langle B_p^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle}\left(\frac{\partial \psi_t}{\partial V} \right)^2 + \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.4)$$

$$\langle B^\theta \rangle = \frac{4\pi^2}{V'}\frac{\partial \psi}{\partial \rho} = 4\pi^2 \frac{\partial \psi}{\partial V} = \frac{I \langle R^{-2} \rangle}{q} = 2\pi \left(\oint \frac{dl_p}{B_p} \right)^{-1} \quad (2.5)$$

$$\hat{q}^2 \equiv \frac{I^2}{2\langle B_p^2 \rangle}\left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \quad (2.6)$$

$$\langle B^2 \rangle \langle R^2 \rangle - I^2 = \langle R^2 \rangle \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 (1 + 2\hat{q}^2) \quad (2.7)$$

$$\langle B \hat{q}_{s\|} \rangle \equiv \frac{2\langle B q_{s\|} \rangle}{5 p_s} \quad (2.8)$$

$$\hat{u}_{s\theta} = \frac{\langle B u_{s\|} \rangle - B V_{1s}}{\langle B^2 \rangle} \quad (2.9)$$

$$\begin{aligned} \left\langle \frac{u_{s\zeta}}{R} \right\rangle &= \frac{1}{1+2\hat{q}^2}\frac{1}{\langle R^2 \rangle}\langle R u_{s\zeta} \rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}\langle B u_{s\|} \rangle \quad (2.10) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + \frac{I}{\langle B^2 \rangle}\left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right)(\langle B u_{s\|} \rangle - B V_{1s}) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + I \left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \hat{u}_{s\theta} \end{aligned}$$

$$\langle R \hat{q}_{s\zeta} \rangle = \frac{1}{\langle B^2 \rangle}\left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right)\frac{1}{I} B V_{2s} + \frac{I}{\langle B^2 \rangle}\langle B \hat{q}_{s\|} \rangle \quad (2.11)$$

$$B V_{1s} = -I \left(\frac{1}{e_s n_s}\frac{\partial n_s T_s}{\partial \psi} + \frac{\partial \Phi}{\partial \psi} \right), \quad B V_{2s} = -\frac{I}{e_s}\frac{\partial T_s}{\partial \psi} \quad (2.12)$$

$$u^\rho \equiv \langle \boldsymbol{u}\cdot\nabla\rho\rangle, \quad u^V = \langle \boldsymbol{u}\cdot\nabla V\rangle \tag{2.13}$$

$$u^\rho_g = -\frac{4\pi^2}{IV'\langle R^{-2}\rangle}\dot{\psi}_t, \quad u^V_g = -\frac{4\pi^2}{I\langle R^{-2}\rangle}\dot{\psi}_t = -\frac{V'}{2\rho\psi_{ta}}\dot{\psi}_t \tag{2.14}$$

$$\frac{1}{V'}\frac{\partial V'}{\partial t} = \frac{\partial}{\partial V}\langle \boldsymbol{u}_g\cdot\nabla V\rangle = \frac{\partial u^V_g}{\partial V} \tag{2.15}$$

$$\langle E_r\rangle = -\langle|\nabla\rho|^2\rangle\frac{\partial r}{\partial\rho}\frac{\partial}{\partial\rho}\Phi \tag{2.16}$$

$$\langle BE_\parallel\rangle = I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) \tag{2.17}$$

$$RE_\zeta = \dot{\psi} = \frac{V_{\rm loop}}{2\pi} \tag{2.18}$$

$$\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\rm i}\rangle = \hat{\mu}^{\rm i}_1(\langle Bu_{\rm i\parallel}\rangle - BV_{1\rm i}) + \hat{\mu}^{\rm i}_2\left(\frac{2\langle Bq_{\rm i\parallel}\rangle}{5p_{\rm i}} - BV_{2\rm i}\right) \tag{2.19}$$

$$\hat{\ell}^{ab}_{ij} \equiv \frac{\ell^{ab}_{ij}}{m_a n_a}, \quad \hat{\ell}^{\rm bb}_{ij} \equiv \frac{\ell^{\rm bb}_{ij}}{m_{\rm e}n_{\rm e}} \tag{2.20}$$

$$\Gamma^{\rm puff} \simeq \frac{\langle \boldsymbol{\Gamma}^{\rm puff}\cdot\nabla\rho\rangle}{\langle|\nabla\rho|\rangle} \tag{2.21}$$

3 Equations in the coordinates suitable for implementation

$$0 = \frac{\partial}{\partial V}\left[\langle|\nabla V|^2\rangle\frac{\partial}{\partial V}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_{\rm e} + Z_{\rm i}en_{\rm i} + Z_{\rm z}en_{\rm z} + Z_{\rm b}en_{\rm b} + Z_{\rm b}eg n^{\rm rp}_{\rm b}\right) \tag{3.1}$$

$$\begin{aligned}\epsilon_0\frac{\partial}{\partial t}\dot{\psi}_t = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\right] - \frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\right]\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\\ & + \frac{1}{\langle B^\theta\rangle}\left[-e\left(n_{\rm e}\langle Bu_{\rm e\parallel}\rangle - In_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle\right) + Z_{\rm i}e\left(n_{\rm i}\langle Bu_{\rm i\parallel}\rangle - In_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle\right)\right.\\ & \left. + Z_{\rm z}e\left(n_{\rm z}\langle Bu_{\rm z\parallel}\rangle - In_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle\right) + Z_{\rm b}e\left(n_{\rm b}\langle Bu_{\rm b\parallel}\rangle - In_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{3.2}$$

$$\begin{aligned}\frac{1}{c^2}\left\langle\frac{1}{R^2}\right\rangle\frac{\partial}{\partial t}\dot{\psi} = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial}{\partial V}\psi\right]\\ & - \mu_0\left(-en_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle + Z_{\rm i}en_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle + Z_{\rm z}en_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle + Z_{\rm b}en_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\end{aligned} \tag{3.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{3.4}$$

$$\frac{\partial}{\partial t}\frac{\psi_t}{\mu_0} = \frac{\dot{\psi}_t}{\mu_0} \tag{3.5}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e} = &-\frac{\partial}{\partial V}n_\mathrm{e}u_\mathrm{e}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{e} + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\ &+ Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e}-n_\mathrm{esup}) + (1-f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{3.6}$$

$$0 = -\frac{e}{m_\mathrm{e}}n_\mathrm{e}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}BV_{1\mathrm{e}} - \frac{e}{m_\mathrm{e}}In_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \frac{e}{m_\mathrm{e}}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle \tag{3.7}$$

$$\begin{aligned}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\|}\rangle = &-\frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}}\\ &+ \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{1}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}T_\mathrm{e} - \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{T_\mathrm{e}}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}n_\mathrm{e}\langle Bu_{\mathrm{i}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}n_\mathrm{e}\langle Bu_{\mathrm{z}\|}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle\\ &- \hat{\ell}_{12}^{\mathrm{ee}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ei}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ez}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{z}\|}\rangle\\ &- e\frac{n_\mathrm{e}}{m_\mathrm{e}}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) - \nu_\mathrm{L}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \frac{\partial}{\partial V}\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &- \frac{\partial}{\partial V}\left[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right.\\ &\left.-\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\partial}{\partial V}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right]\end{aligned} \tag{3.8}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle = &-(u_\mathrm{e}^V-u_g^V)\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial u_\mathrm{e}^V}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle]\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\langle Ru_{\mathrm{e}\zeta}\rangle\frac{\partial}{\partial V}n_\mathrm{e} - \frac{1}{m_\mathrm{e}}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}\frac{n_\mathrm{e}}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}\frac{n_\mathrm{e}}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}_{12}^{\mathrm{ee}} n_{\mathrm{e}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ee}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ei}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{ei}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- e \frac{n_{\mathrm{e}}}{m_{\mathrm{e}}} \dot{\psi} - e \frac{1}{m_{\mathrm{e}}} \frac{\partial \psi}{\partial V} n_{\mathrm{e}} u_{\mathrm{e}}^V \\
&- \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{e}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[n_{\mathrm{e}} \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{e}} \frac{\partial}{\partial V} \Phi + \frac{C_{\mathrm{T}} - 1}{e} \left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{e}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle
\end{aligned} \tag{3.9}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{e}} T_{\mathrm{e}} = &- \frac{5}{2} (u_{\mathrm{e}}^V - u_g^V) \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{5}{2} \frac{\partial u_{\mathrm{e}}^V}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} [V_{\mathrm{e}}^{\mathrm{hp}} \langle |\nabla V| \rangle n_{\mathrm{e}} T_{\mathrm{e}}] \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- u_g^V \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - u_{\mathrm{i}}^V \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - u_{\mathrm{z}}^V \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle \\
&- \frac{3}{2} \nu_{\mathrm{Tei}} \left(n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{Tez}} \left(n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- 4\pi^2 \frac{\partial \psi}{\partial V} \dot{\psi}_t (-e n_{\mathrm{e}} \hat{u}_{\mathrm{e}\theta} + Z_{\mathrm{i}} e n_{\mathrm{i}} \hat{u}_{\mathrm{i}\theta} + Z_{\mathrm{z}} e n_{\mathrm{z}} \hat{u}_{\mathrm{z}\theta}) \\
&+ \dot{\psi} \left(-e n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + Z_{\mathrm{i}} e n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + Z_{\mathrm{z}} e n_{\mathrm{z}} \left\langle \frac{u_{\mathrm{z}\zeta}}{R} \right\rangle \right) \\
&- \nu_{\mathrm{L}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{esup}} T_{\mathrm{e}}) - \frac{3}{2} \nu_{\mathrm{LT_e}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} T_{\mathrm{esup}}) - E_{\mathrm{H}} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\
&+ f_{\mathrm{ei}} P_{\mathrm{b}} + P_{\alpha \mathrm{e}} + P_{\mathrm{RFe}} - P_{\mathrm{Br}}
\end{aligned} \tag{3.10}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{e}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle = &- \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \frac{1}{m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} B V_{1\mathrm{e}} \\
&+ \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{e}}}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- \hat{\ell}_{21}^{\mathrm{ee}} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ei}} n_{\mathrm{e}} \langle B u_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ez}} n_{\mathrm{e}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{eb}} \frac{n_{\mathrm{e}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ee}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ei}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ez}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned} \tag{3.11}
$$

$$
0 = - n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle \tag{3.12}
$$

$$0 = -n_{\mathrm{e}} B V_{1\mathrm{e}} + \frac{1}{e}\frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.13}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} = &-\frac{\partial}{\partial V} n_{\mathrm{i}} u_{\mathrm{i}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{i}} + \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) - \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) \\ &+ \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}\end{aligned} \tag{3.14}$$

$$0 = Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{i}} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} I n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \langle B^2 \rangle n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \tag{3.15}$$

$$\begin{aligned}n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B u_{\mathrm{i}\parallel} \rangle = &-\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{i}} \\ &- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} n_{\mathrm{i}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} n_{\mathrm{i}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\ &- \hat{\ell}_{12}^{\mathrm{ie}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{ii}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{iz}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\ &+ Z_{\mathrm{i}} e \frac{n_{\mathrm{i}}}{m_{\mathrm{i}}} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\ &- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{LC}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &- \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right. \\ &\left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right]\end{aligned} \tag{3.16}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle = &-(u_{\mathrm{i}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial u_{\mathrm{i}}^{V}}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \langle R u_{\mathrm{i}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} - \frac{1}{m_{\mathrm{i}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} \frac{n_{\mathrm{i}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle \\
&+ \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle \\
&+ \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&+ Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \dot{\psi} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \frac{\partial \psi}{\partial V} n_{\mathrm{i}} u_{\mathrm{i}}^{V} \\
&+ Z_{\mathrm{i}} \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{i}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[(n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + (n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial}{\partial V} \Phi \right. \\
&\left. + \frac{C_{\mathrm{T}} - 1}{e} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{i}} + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{b}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle
\end{aligned} \tag{3.17}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{i}} T_{\mathrm{i}} = &- \frac{5}{2} (u_{\mathrm{i}}^{V} - u_{g}^{V}) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{5}{2} \frac{\partial u_{\mathrm{i}}^{V}}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} V_{\mathrm{i}}^{\mathrm{hp}} \langle |\nabla V| \rangle n_{\mathrm{i}} T_{\mathrm{i}} \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle T_{\mathrm{i}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&+ (u_{\mathrm{i}}^{V} - u_{g}^{V}) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} \nu_{\mathrm{Tei}} \left(\frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{e}} T_{\mathrm{e}} \right) - \frac{3}{2} \nu_{\mathrm{Tiz}} \left(n_{\mathrm{i}} T_{\mathrm{i}} - \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- \nu_{\mathrm{L}} \left(n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{isup}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{LT_i}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{1}{n_{\mathrm{b}}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \frac{\langle B u_{\mathrm{i}\parallel} \rangle}{\langle B^2 \rangle} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle + P_{\alpha \mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{3.18}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle = &- \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_3^{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} B V_{1\mathrm{i}} \\
&- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_3^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_3^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&- \hat{\ell}^{\mathrm{ie}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{e}\parallel} \rangle - \hat{\ell}^{\mathrm{ii}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \hat{\ell}^{\mathrm{iz}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\ell}^{\mathrm{ib}}_{21} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \hat{\ell}^{\mathrm{ie}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \hat{\ell}^{\mathrm{ii}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \hat{\ell}^{\mathrm{iz}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle
\end{aligned} \tag{3.19}
$$

$$
0 = - n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle \tag{3.20}
$$

$$0 = -n_{\mathrm{i}} BV_{1\mathrm{i}} - \frac{1}{Z_{\mathrm{i}} e}\frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.21}$$

$$\frac{\partial}{\partial t} n_{\mathrm{z}} = -\frac{\partial}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{z}} + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 - \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{isup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}} \tag{3.22}$$

$$0 = Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} BV_{1\mathrm{z}} + Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} I n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} \langle B^2 \rangle n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \tag{3.23}$$

$$\begin{aligned}
n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel} \rangle = &-\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \frac{1}{m_{\mathrm{z}}} \hat{\mu}_1^{\mathrm{i}} BV_{1\mathrm{z}} \\
&- \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{z}}}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} \\
&+ \hat{\ell}_{11}^{\mathrm{ze}} n_{\mathrm{z}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} n_{\mathrm{z}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&- \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&+ Z_{\mathrm{z}} e \frac{n_{\mathrm{z}}}{m_{\mathrm{z}}} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{z}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
&- \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} BV_{1\mathrm{z}} \right. \\
&\left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} BV_{1\mathrm{z}} \right]
\end{aligned} \tag{3.24}$$

$$\begin{aligned}
\frac{\partial}{\partial t} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle = &-(u_{\mathrm{z}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \\
&+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \langle R u_{\mathrm{z}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{z}} - \frac{1}{m_{\mathrm{z}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
&+ \hat{\ell}_{11}^{\mathrm{ze}} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} \frac{n_{\mathrm{z}}}{n_{\mathrm{i}}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\
&- \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned}$$

$$
\begin{aligned}
&+\hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{i}\|}\rangle \\
&+\hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}}\dot{\psi} + Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}}\frac{\partial \psi}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} \\
&+ Z_{\mathrm{z}}\frac{e\langle|\nabla V|^2\rangle D_{\mathrm{e}}}{m_{\mathrm{z}} T_{\mathrm{e}}}\frac{\partial \psi}{\partial V}\left[n_{\mathrm{z}}\frac{\partial \psi}{\partial V}\frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{z}}\frac{\partial}{\partial V}\Phi\right. \\
&\left.+\frac{C_{\mathrm{T}}-1}{e}\frac{n_{\mathrm{z}}}{n_{\mathrm{e}}}\left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}}\frac{\partial}{\partial V} n_{\mathrm{e}}\right) + \frac{T_{\mathrm{e}}}{e}\frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{z}}\right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{3.25}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{\partial}{\partial t} n_{\mathrm{z}} T_{\mathrm{z}} &= -\frac{5}{2}(u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{5}{2}\frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V} V_{\mathrm{z}}^{\mathrm{hp}}\langle|\nabla V|\rangle n_{\mathrm{z}} T_{\mathrm{z}} \\
&+ \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle T_{\mathrm{z}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&+ (u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta}\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle - \frac{3}{2}\nu_{\mathrm{Tez}}\left(\frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{e}} T_{\mathrm{e}}\right) - \frac{3}{2}\nu_{\mathrm{Tiz}}\left(\frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{i}} T_{\mathrm{i}}\right) \\
&- \nu_{\mathrm{L}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{zsup}} T_{\mathrm{z}}\right) - \frac{3}{2}\nu_{\mathrm{LT_z}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}} T_{\mathrm{zsup}}\right) + P_{\alpha\mathrm{z}} + P_{\mathrm{RFz}}
\end{aligned} \tag{3.26}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{z}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{z}\|}\rangle &= -\frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \frac{1}{m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle + \frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} \\
&- \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{1}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{T_{\mathrm{z}}}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&- \hat{\ell}_{21}^{\mathrm{ze}} n_{\mathrm{z}}\langle B u_{\mathrm{e}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zi}} n_{\mathrm{z}}\langle B u_{\mathrm{i}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zz}} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zb}}\frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}}\langle B u_{\mathrm{b}\|}\rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ze}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zi}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zz}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle
\end{aligned} \tag{3.27}
$$

$$
0 = -n_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle \tag{3.28}
$$

$$
0 = -n_{\mathrm{z}} B V_{1\mathrm{z}} - \frac{1}{Z_{\mathrm{z}} e}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V}\Phi \tag{3.29}
$$

$$\frac{\partial}{\partial t}n_\mathrm{b} = -\frac{\partial}{\partial V}n_\mathrm{b}u_\mathrm{b}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{b} + \frac{P_\mathrm{b}}{E_\mathrm{b}} - g\frac{P_{\mathrm{b}\perp}}{E_\mathrm{b}} - \nu_\mathrm{b}n_\mathrm{b} - \nu_\mathrm{bL}n_\mathrm{b} - \nu_\mathrm{eff}g(\sqrt{\delta}n_\mathrm{b} - n_\mathrm{b}^\mathrm{rp})$$
$$+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.30}$$

$$0 = Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}n_\mathrm{b}BV_{1\mathrm{b}} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}In_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\langle B^2\rangle n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle \tag{3.31}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle = -\frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}BV_{1\mathrm{b}}$$
$$+ \hat{\ell}_{11}^{\mathrm{bb}}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \sum_{a\neq\mathrm{b}}\hat{\ell}_{11}^{\mathrm{b}a}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}}\langle Bu_{a\|}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q} + \dot{\psi}\right) - \nu_\mathrm{bL}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle$$
$$- \nu_\mathrm{CX}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{1}{m_\mathrm{b}}\langle BS_{\mathrm{m}\|\mathrm{b}}\rangle - \nu_\mathrm{eff}g\sqrt{\delta}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\langle Bu_{\mathrm{b}\|}\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.32}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle = u_g^V\frac{\partial}{\partial V}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{\partial}{\partial V}\langle Ru_{\mathrm{b}\zeta}\rangle n_\mathrm{b}u_\mathrm{b}^V$$
$$+ \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{be}}\frac{1}{n_\mathrm{e}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bi}}\frac{1}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bz}}\frac{1}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bb}}\frac{1}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}\dot{\psi} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\frac{\partial\psi}{\partial V}n_\mathrm{b}u_\mathrm{b}^V$$
$$- \nu_\mathrm{bL}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{CX}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\frac{I}{\langle B^2\rangle}\langle BS_{\mathrm{m}\|\mathrm{b}}\rangle \tag{3.33}$$

$$0 = -n_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle \tag{3.34}$$

$$0 = -n_\mathrm{b}BV_{1\mathrm{b}} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}In_\mathrm{b}\frac{\partial}{\partial V}T_\mathrm{b} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}IT_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} - n_\mathrm{b}\frac{\partial V}{\partial\psi}I\frac{\partial}{\partial V}\Phi \tag{3.35}$$

$$\frac{\partial}{\partial t}n_{01} = -\frac{\partial u_g^V}{\partial V}n_{01} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{01}\frac{\partial}{\partial V}n_{01} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{01}$$
$$- \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} + \gamma_0\nu_\mathrm{L}(n_\mathrm{i} - n_\mathrm{isup}) + \frac{\partial}{\partial V}V'\Gamma^\mathrm{puff}\Big|_\mathrm{b} \tag{3.36}$$

$$\frac{\partial}{\partial t}n_{02} = -\frac{\partial u_g^V}{\partial V}n_{02} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{02}\frac{\partial}{\partial V}n_{02} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{02} + \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} \tag{3.37}$$

$$\frac{\partial}{\partial t}n_{03} = -\frac{\partial u_g^V}{\partial V}n_{03} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{03}\frac{\partial}{\partial V}n_{03} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{03} + f_\mathrm{CX}\frac{P_\mathrm{b}}{E_\mathrm{b}} \tag{3.38}$$

$$\frac{\partial}{\partial t}\bar{n}_z = -\frac{\partial u_g^V}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{nz}\frac{\partial}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle \Delta D_{\mathrm{C0}}\frac{\partial}{\partial V}n_{\mathrm{C0}} - n_{\mathrm{e}}\bar{n}_z f_5 S_{\bar{5}} + n_{\mathrm{e}} n_{\mathrm{z}} A_6$$
$$+ \gamma_{0\mathrm{z}}\nu_{\mathrm{L}}\left(n_{\mathrm{z}} - n_{\mathrm{zsup}}\right) + \gamma_{\mathrm{i,z}}\nu_{\mathrm{L}}\left(n_{\mathrm{i}} - n_{\mathrm{isup}}\right) - (1-\gamma_{0\mathrm{z}})\nu_{\mathrm{L}}\left(\bar{n}_z - n_{\mathrm{C0}}\right)$$
$$+ \frac{\partial}{\partial V}V'\Gamma_{\mathrm{z}}^{\mathrm{puff}}\Big|_{\mathrm{b}} \tag{3.39}$$

$$\frac{\partial}{\partial t}n_{\mathrm{b}}^{\mathrm{rp}} = -\frac{\partial u_g^V}{\partial V}n_{\mathrm{b}}^{\mathrm{rp}}\frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} + \nu_{\mathrm{eff}}(\sqrt{\delta}n_{\mathrm{b}} - n_{\mathrm{b}}^{\mathrm{rp}}) - \nu_{\mathrm{b}}n_{\mathrm{b}}^{\mathrm{rp}}$$
$$- \frac{\partial}{\partial V}\langle|\nabla V|\rangle u_{\mathrm{b}}^{\mathrm{rp}}n_{\mathrm{b}}^{\mathrm{rp}} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{\mathrm{b}}^{\mathrm{rp}}\frac{\partial}{\partial V}n_{\mathrm{b}}^{\mathrm{rp}} \tag{3.40}$$

4 Boundary Conditions

num.	name	variable	highest order derivative	center	edge
1	LQm1	Φ	2	N	0
2	LQm2	$\dot{\psi}_t$	0	×	×
3	LQm3	$\dot{\psi}$	0	×	×
4	LQm4	ψ	2	N	$2\pi\mu_0 I_\mathrm{p}$
5	LQm5	ψ_t/μ_0	2	0	$\langle\lvert\nabla V\rvert^2/R^2\rangle\lvert_{V_b} I_\mathrm{vac}/(4\pi^2)$
6	LQe1	n_e	2	N	N†
7	LQe2	$n_\mathrm{e}u_\mathrm{e}^V$	1	0	×
8	LQe3	$\langle Bu_{\mathrm{e}\parallel}\rangle$	2(0)	N(×)	N(×)
9	LQe4	$n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle$	2	N	N
10	LQe5	$n_\mathrm{e}T_\mathrm{e}$	2	N	N
11	LQe6	$\langle B\hat{q}_{\mathrm{e}\parallel}\rangle$	0	×	×
12	LQe7	$n_\mathrm{e}\langle u_{\mathrm{e}\zeta}/R\rangle$	0	×	×
13	LQe8	$BV_{1\mathrm{e}}$	0	×	×
14	LQi1	n_i	2	N	N†
15	LQi2	$n_\mathrm{i}u_\mathrm{i}^V$	1	0	×
16	LQi3	$\langle Bu_{\mathrm{i}\parallel}\rangle$	2(0)	N(×)	N(×)
17	LQi4	$n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle$	2	N	N
18	LQi5	$n_\mathrm{i}T_\mathrm{i}$	2	N	N
19	LQi6	$\langle B\hat{q}_{\mathrm{i}\parallel}\rangle$	0	×	×
20	LQi7	$n_\mathrm{i}\langle u_{\mathrm{i}\zeta}/R\rangle$	0	×	×
21	LQi8	$BV_{1\mathrm{i}}$	0	×	×
22	LQz1	n_z	2	N	N†
23	LQz2	$n_\mathrm{z}u_\mathrm{z}^V$	1	0	×
24	LQz3	$\langle Bu_{\mathrm{z}\parallel}\rangle$	2(0)	N(×)	N(×)
25	LQz4	$n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle$	2	N	N
26	LQz5	$n_\mathrm{z}T_\mathrm{z}$	2	N	N
27	LQz6	$\langle B\hat{q}_{\mathrm{z}\parallel}\rangle$	0	×	×
28	LQz7	$n_\mathrm{z}\langle u_{\mathrm{z}\zeta}/R\rangle$	0	×	×
29	LQz8	$BV_{1\mathrm{z}}$	0	×	×
30	LQb1	n_b	0(2)	×(N)	×(N)
31	LQb2	$n_\mathrm{b}u_\mathrm{b}^V$	1	0	×
32	LQb3	$n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle$	0	×	×
33	LQb4	$n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$	0	×	×
34	LQb7	$n_\mathrm{b}\langle u_{\mathrm{b}\zeta}/R\rangle$	0	×	×
35	LQb8	$BV_{1\mathrm{b}}$	0	×	×
36	LQn1	n_{01}	2	N	$\langle\lvert\nabla V\rvert\rangle S_\mathrm{gas}\rvert_b$
37	LQn2	n_{02}	2	N	0
38	LQn3	n_{03}	2	N	0
39	LQnz	$\bar{n}_z$	2	N	$\langle\lvert\nabla V\rvert\rangle S_{\mathrm{gas,z}}\rvert_b$
40	LQr1	n_b^rp	0(2)	×(N)	×(N)

0: Dirichlet condition ($u = 0$), N: Neumann condition ($u' = 0$)

value: Neumann condition ($u' =$ value), ×: no condition

†: imposed as a constraint in V5.60 (`MDLWBC` = 1), see Sec. 5.2

NOTE: The dependent variable of LQm4 turns to be ψ/μ_0 when NB is applied. Subsequently, the edge boundary condition listed for that variable is also changed to $2\pi I_\mathrm{p}$.

NOTE: Rows 2 to 5 deserve a word, because the variable that carries the second-order operator is not the variable of the equation in which the operator is written. The LQm2 equation advances $\dot{\psi}_t$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \langle R^{-2}\rangle^{-1} \frac{\partial}{\partial V}(\psi_t/\mu_0)]$; the LQm3 equation advances $\dot{\psi}$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \frac{\partial}{\partial V}\psi]$. Counting per dependent variable over the whole system, as this section does, $\dot{\psi}_t$ and $\dot{\psi}$ are never differentiated and so take order 0 and no condition of their own, while ψ_t/μ_0 and ψ take order 2 and carry the two conditions each. The table above now reads that way. In the implementation the two conditions of each pair are applied to the LQm2 and LQm3 equation rows, since that is where the boundary term left by the integration by parts appears; this is the correct weak form and is not a discrepancy with the table. The axis condition of ψ_t/μ_0 is imposed there as $\dot{\psi}_t|_{V=0} = 0$, which holds $\psi_t(0)$ at its initial value, namely zero, and is therefore the Dirichlet condition listed.

NOTE: In the table the column labeled by the highest order derivative means the order of the highest order derivative of a dependent variable, not an equation, in the corresponding row. Please note that the term not integrated by parts is never counted. For example, the first row shows the highest order of derivatives is equal to 2 for Φ because the LQm1 equation has the diffusion term having the second derivative of Φ with respect to V. In addition, the LQe2 and LQi2 equations also have the first derivative of Φ, which is not a dependent variable of these equations. As a whole, the highest order derivative of Φ in this equation system is 2 that comes from the LQm1 equation. When we look at n_e, we find that the highest order derivative of n_e is 2 shown in the LQe5 equation, even though no derivative appears in the LQe1 equation. The Neumann conditions for n_e come from the LQe5 equation. When it comes to $n_\mathrm{e}u_\mathrm{e}^V$, the highest derivative is found as order 1 in the LQe1 equation, but this term is not integrated by parts. It means that any boundary condition is not naturally imposed on $n_\mathrm{e}u_\mathrm{e}^V$. The Dirichlet boundary condition should be imposed at axis.

5 Models, switches and defaults of V5.60

V5.60 changes the default SOL and neutral physics. The detailed physics, the verification and the development history are in `neutral_transport.tex`; this section states what the code solves.

5.1 SOL parallel loss

Outside the separatrix the parallel loss rate is

$$\nu_{\rm L} = k_L \frac{C_s}{L_c}\, g(\rho), \qquad L_c = 2\pi q R, \qquad g = \frac{x^2}{1+x^2}, \quad x = \frac{r-a}{\delta}\ (r \ge a),\ x = 0\ (r < a), \tag{5.1}$$

and enters every SOL equation in Robin form, e.g. $-\nu_{\rm L}(n_s - n_{\rm ssup})$. k_L is `FSLP`. The ramp $g(\rho)$ is numerical smoothing: it keeps $\nu_{\rm L}$ from jumping at $\rho = 1$, which the FEM cannot represent, and its continuum limit is the step. `MDLSOL` selects its width δ:

`MDLSOL`	δ
0	$3\sqrt{0.5\,T_i(0)\,m_i}\,q(a)/(Z_i e B_\phi(a))/\sqrt{a/R}$ (V5.52)
1	ion banana width at the separatrix, $\sqrt{\epsilon}\,\sqrt{T_i m_i}/(Z_i e B_\theta)$ (default)

With `MDLSOL` = 0 the width is evaluated with half the on-axis ion temperature and the edge q; for the X-point test equilibrium distributed with the code (`eqdata`, $R = 3.2$ m, $a = 0.8$ m) it is 1.8 times the whole SOL, so g is only 0.17 at the wall and the ramp suppresses the loss everywhere instead of smoothing its onset. `MDLSOL` = 1 evaluates the banana width with the local ion temperature and poloidal field at the separatrix and gives $g \simeq 0.99$ at the wall.

The default $k_L = 1$ is the value of the sheath-limited two-point model (pressure conservation, Mach number 1 at the target, upstream and target densities related by $n_{\rm t} = n_{\rm u}/2$) when $2\pi qR$ is read as the length between the two targets: a flux tube of that length loses $2n_{\rm t}C_s$ per unit area while holding $n_{\rm u}\,2\pi qR$, so $\nu_{\rm L} = C_s/(2\pi qR)$. V5.52 used the empirical value $k_L = 0.3$, a typical upstream Mach number of the SOL flow; with `MDLSOL` = 1 it does not give a stable SOL on the X-point test equilibrium.

5.2 Wall boundary condition for the densities

The table of the previous section lists a Neumann condition for the densities at the wall $\rho = \rho_b$, said to come naturally from the pressure equation. The natural boundary condition of that equation is, however, zero heat flux, not a condition on the density. Up to V5.52 the densities therefore had no condition at the wall, and the radial particle flux through ρ_b was fixed by the last element. On the X-point test equilibrium the ratio of the ion flux through the wall to that through the separatrix, $\Gamma_{\rm wall}/\Gamma_{\rm sep}$, was $+0.57$, -0.40 and -0.28 at `NRMAX` = 60, 120 and 480: depending on the mesh the wall acted as a source or a sink of particles, and refining the mesh did not make it vanish. The radial flux $n_s u_{\rm s}^V$ cannot take the condition instead: its equation is first order and not integrated by parts, so it already has its one condition at the axis. For the charged species ρ_b is a computational boundary at the outer edge of the SOL, not a material wall: their exhaust is carried by $\nu_{\rm L}$.

V5.60 (`MDLWBC` = 1) imposes $dn_s/d\rho = 0$ on $n_{\rm e}$, $n_{\rm i}$ and $n_{\rm z}$. In the P1 basis this is the constraint

$$n_s(N) = n_s(N-1), \qquad N = \texttt{NRMAX}. \tag{5.2}$$

Constrained degrees of freedom, in general. For a linear system $Ax = b$ with the constraint $x_N = x_{N-1}$, write $x = Ty$, where T is the identity on $y_1 \dots y_{N-1}$ and copies y_{N-1} into x_N. The constrained Galerkin system is

$$T^{\mathrm{T}} A T\, y = T^{\mathrm{T}} b. \tag{5.3}$$

T^{T} adds row N to row $N-1$; AT adds column N to column $N-1$. In finite elements this is the basis reduction $\phi_{N-1} \to \phi_{N-1} + \phi_N$: the trial and test functions are constant on the last element, which is the P1 form of $dn/d\rho = 0$. The code keeps the system square and solves the equivalent

$$\text{row } N-1 \leftarrow \text{row } N-1 + \text{row } N, \qquad \text{row } N \leftarrow x_N - x_{N-1} = 0, \tag{5.4}$$

with x_N still an unknown, so the other equations keep their coupling to it and the column sum is carried by the constraint row. Since $\sum_i \phi_i = 1$, the sum of all rows of a continuity equation is the global particle balance; each column of T^{T} sums to one, so that sum, and with it particle conservation, is unchanged. Replacing row N by the constraint without adding it to row $N-1$ would drop the particle balance of the wall half-cell, an $O(h)$ conservation error.

Implementation. `txcala.f90`, subroutine `BOUNDARY_ZEROGRAD(LQ)`, called from the boundary-condition block of `TXCALA` for `LQe1`, `LQi1` and `LQz1` after the Dirichlet eliminations. In the band storage the coefficient of term `NC` of equation `NQ` in row `NR` is `ALC`, `BLC` or `CLC(NR,NC,NQ)` for `X(NR+1)`, `X(NR)` or `X(NR-1)`, and `PLC` holds the known terms. For every term `NC = 0..NLCMAX`: `BLC(N-1) += CLC(N)`, `ALC(N-1) += BLC(N)`, `PLC(N-1) += PLC(N)`; row `N` is then cleared and set to `BLC(N,1) = 1`, `CLC(N,1) = -1`. The time term and its right-hand side are built from the same coefficient arrays, so they are folded with them. The condition converges at first order in the mesh size: $\Gamma_{\mathrm{wall}}/\Gamma_{\mathrm{sep}}$ is -1.8, -1.1, -0.5 and -0.3% at `NRMAX` = 60, 120, 240 and 480.

5.3 C^0–C^{5+} superstage with a source-slaved C^0 part

Carbon is carried by two densities: n_{z} for C^{6+} and the superstage $\bar{n}_z = \sum_{q=0}^{5} n_{C^{q+}}$, called the bundle below. The bundle exchanges particles with C^{6+} through the effective ionization $f_5 S_5$ and recombination A_6 of Sec. 1, and is fed by sputtering in the SOL: $\gamma_{0\mathrm{z}}$ (`rGamm0zz`) is the fraction of the C^{6+} parallel loss and $\gamma_{\mathrm{i,z}}$ (`rGamm0iz`) the fraction of the main-ion parallel loss that return as sputtered carbon.

In local ionization balance the C^0 fraction of the bundle is below 2×10^{-6}, so the bundle is almost entirely magnetized C^{1+}–C^{5+} ions. With `MDLNZD` = 1 it is therefore diffused with the anomalous particle diffusivity of the charged species, $D_{n\mathrm{z}} = D_{\mathrm{c}}$ (`Dfs`). The exception is the carbon that has just been sputtered, which enters as neutral C^0. Its lifetime $1/(S_0 n_{\mathrm{e}})$ is of the order of microseconds, shorter than every other time scale here, so its density follows its local source balance:

$$n_{\mathrm{C0}} = \min\left(\frac{S_{\mathrm{wall}}}{S_0 n_{\mathrm{e}}},\ \bar{n}_z\right), \quad S_{\mathrm{wall}} = \gamma_{0\mathrm{z}} \nu_{\mathrm{L}} \max(n_{\mathrm{z}} - n_{\mathrm{zsup}}, 0) + \gamma_{\mathrm{i,z}} \nu_{\mathrm{L}} \max(n_{\mathrm{i}} - n_{\mathrm{isup}}, 0), \tag{5.5}$$

where S_0 is the $C^0 \to C^{1+}$ ionization rate (OpenADAS scd96, `data/Carbon-S0F0_20260921.txt`). n_{C0} enters the $\bar{n}_z$ equation as a known field through the excess flux with $\Delta D_{\mathrm{C0}} = D_{\mathrm{C0}} - D_{\mathrm{c}}$, the part of the C^0 flux $-D_{\mathrm{C0}} \nabla n_{\mathrm{C0}}$ that the D_{c} term does not already carry, so that the bundle flux is exactly $-D_{\mathrm{c}} \nabla(\bar{n}_z - n_{\mathrm{C0}}) - D_{\mathrm{C0}} \nabla n_{\mathrm{C0}}$. Here D_{C0} is the Larsen-limited (see below) value of $\bar{v}_{0z}^2/(3 S_0 n_{\mathrm{e}})$, the short-mean-free-path diffusivity of C^0, and $\bar{v}_{0z} = \sqrt{4 V_{0z}^2/\pi}$ its mean speed (V_{0z} = `V0z`), and through the parallel loss of the charged part of the bundle, $(1 - \gamma_{0\mathrm{z}}) \nu_{\mathrm{L}} (\bar{n}_z - n_{\mathrm{C0}})$, of which the fraction $\gamma_{0\mathrm{z}}$ is sputtered back (the $\bar{n}_z$ equations of Secs. 1 and 3). n_{C0} is

evaluated from the previous Picard iterate, $S_{\rm wall}$, $S_0 n_{\rm e}$ and the bound $\bar{n}_z$ alike, so the bound $n_{\rm C0} \le \bar{n}_z$ applies to the bundle at the time level being solved for, and the $\bar{n}_z$ equation stays linear in $\bar{n}_z$. With `MDLNZD` $= 0$ (the V5.52 model), $D_{nz} = D_{01}$, the diffusivity of the slow hydrogen neutrals, and the $n_{\rm C0}$ terms are absent.

Element-based C^0 limiter. $D_{\rm C0}$ is Larsen-limited, $D \to D/(1 + D|\nabla n_{\rm C0}|/(\alpha \bar{v}_{0z} n_{\rm C0}))$, $\alpha =$ `FLlim`. The limiter is evaluated on each element from the element mean of $n_{\rm C0}$ and the element gradient. It acts on the whole $D_{\rm C0}$, because it bounds the whole C^0 flux by free streaming, and $D_{\rm c}$ is subtracted afterwards with its sign: $\Delta D_e = D_e - D_{{\rm c},e}$, where D_e is the limited $D_{\rm C0}$ and $D_{{\rm c},e}$ the element mean of $D_{\rm c}$. $\Delta D_e < 0$ is not a negative diffusion: it removes the part $D_{{\rm c},e}$ of the C^0 flux that the $D_{\rm c}$ term already carries, and since $n_{\rm C0}$ is a known field the matrix is unchanged. It occurs in the first element outside the separatrix, where $n_{\rm C0}$ rises from zero within one element and the limited $D_{\rm C0}$ falls to about $\alpha \bar{v}_{0z} h/2$ (h the radial width of the element): at NRMAX $= 480$, not at NRMAX ≤ 240. Clipping ΔD_e at zero there would let the C^0 flux exceed free streaming by 7%. The element flux $F_e = \langle |\nabla V|^2 \rangle_e \Delta D_e \, \Delta n_{\rm C0}/\Delta V_e$ enters the two rows of the element as $+F_e$ and $-F_e$, so the term is exactly conservative. The element evaluation is needed because inside the separatrix the source, and hence $n_{\rm C0}$, vanishes: a nodal ratio $|\nabla n_{\rm C0}|/n_{\rm C0}$ would be $0/0$ at the separatrix node, and whether that node is limited would be left to rounding. The limiter is part of the model and is always applied with `MDLNZD` $= 1$. Without it $D_{\rm C0}$ becomes very large in the SOL, and the divergence of the known C^0 flux, a source that does not scale with $\bar{n}_z$, drives $\bar{n}_z$ negative just outside the separatrix.

Validity of the source-slaved C^0 part. The slaving of C^0 to its source is implemented and is part of the production model. What remains open is the range of validity of this quasi-steady approximation and the lack of a formal positivity guarantee. The mean free path of C^0 is comparable to the width of the C^0 layer, so the local source balance does not resolve the inner structure of the layer. Positivity of $\bar{n}_z$ is kept by the limiter in the cases computed, but it is not guaranteed, because the divergence of the known C^0 flux is a source that does not scale with $\bar{n}_z$. The bound $n_{\rm C0} = \bar{n}_z$ marks the limit of the model. It is never reached in the converged iterates of the production cases; it is reached only in the first sweep of a run started from the initial profiles, where $\bar{n}_z$ is still close to zero. Where it holds persistently, the lagged treatment of $n_{\rm C0}$ is not appropriate: the C^0 flux in the bounded nodes is then a diffusion of $\bar{n}_z$ itself with $\Delta D_{\rm C0} \approx 7$–$15\, D_{\rm c}$, taken from the previous iterate, and the iteration diverges (forced with a C^0 ionization rate reduced 100-fold; with the bound taken from the previous time step, the time stepping diverges instead). `TXLOOP` warns when the bound is still active in the last Picard sweep of a step, and with `IDIAG` ≥ 2 prints the number of bounded nodes at every step. Such conditions call for an independent C^0 transport equation, an implicit treatment of the C^0 flux, or a kinetic (Monte-Carlo) treatment, which would also delimit the validity of the approximation.

5.4 Larsen limiter on the hydrogen slow neutrals (optional)

The slow hydrogen neutrals are diffused with the short-mean-free-path diffusivity $D_{01} = \bar{v}_0^2/(3 S_{\rm tot}) = \lambda \bar{v}_0/3$, $S_{\rm tot}$ being the ionization plus charge-exchange rate and $\lambda = \bar{v}_0/S_{\rm tot}$ the mean free path. The Larsen limiter replaces D by $D/(1+R)$ with

$$R = \frac{D|\nabla n_{01}|}{\alpha\, \bar{v}_0\, n_{01}} = \frac{\text{flux the diffusion law demands}}{\text{free-streaming flux it may not exceed}} = \frac{\lambda}{3\alpha L_n}, \qquad \alpha = \texttt{FLlim},$$

L_n being the gradient length: the limited flux $D|\nabla n|/(1+R)$ is the diffusion flux where $R \ll 1$ and $\alpha \bar{v}_0 n$ where $R \gg 1$. $\alpha = 1/4$ is the one-sided flux of an isotropic population: for particles of speed $\bar{v}$ spread uniformly over

all directions, those crossing a plane in one direction carry $n\bar{v}\int_0^1 \mu\, d\mu/2 = n\bar{v}/4$. $\alpha = 1/2$ would correspond to all particles moving into one half-space, uniformly in angle, and $\alpha = 1$ to all of them moving in the same direction, the kinematic upper bound. In the V5.60 solution without the limiter R is 1–3 in the SOL, about 2 at the separatrix and up to 6 at $\rho \simeq 0.8$–0.9 (Kn $= \lambda/L_n \sim 1$): the diffusion approximation is marginal wherever the neutrals matter.

`MDLNOFL` $= 0$ (default) leaves D_{01} unlimited. `MDLNOFL` $= 2$ evaluates the limiter element by element, as for the C^0 flux, from the element mean $\bar{n}$ and the element difference Δn of the current iterate; since $|\Delta n|/\bar{n} \le 2$ on an element, the factor is bounded by $1 + 2\lambda/(3\alpha h)$. `MDLNOFL` $= 1$ is the earlier nodal form, with $|\nabla n|/n$ from the spline derivative, kept only to reproduce earlier runs. The limited ionization front at $\rho \simeq 0.85$–0.9 is exponential with the decay length $\alpha\lambda \simeq 5$ mm, shorter than an element at `NRMAX` $= 60$ (about 1 cm). On that unresolved front, and in the tiny-n_{01} tail inside it, the nodal ratio is large and erratic, and the Picard iteration does not settle: with `MDLNOFL` $= 1$, 496 of 500 steps reach `ICMAX` at `NRMAX` $= 60$ (`EPS` $= 10^{-6}$), the neutral-only relaxation `RLXNO` $= 0.5$ (used only with `MDLNOFL` $= 1$) leaves 2–3. `MDLNOFL` $= 2$ needs no relaxation: 20.0 / 21.7 sweeps on average and no step at `ICMAX` (X-point / circular, `NRMAX` $= 60$), the same as without the limiter. The two forms agree to 0.5–2% where both converge. Against the limiter off, the limiter raises the neutral inventory by about 30% and changes n_e at the separatrix and at the wall by -3% and $+4\%$; $Z_{\rm eff}(0)$ is unchanged.

The limiter is a bound on the diffusion flux, not a substitute for kinetic neutral transport. A straight-flight reference (the same sources, attenuated at $S_{\rm tot}$) ionizes or charge-exchanges 35% of the slow neutrals inside the separatrix (X-point, `NRMAX` $= 240$); plain diffusion gives 49% and the limiter with $\alpha = 1/4$ 25%, with almost nothing beyond $\rho = 0.95$ against 12% in the reference. $\alpha = 1/2$ recovers the total but not the ballistic tail, and no single α recovers both. The difference between limiter on and off is of the size of the uncertainty from the neutral energy alone (the reference gives 24% and 48% at 0.5 and 4.5 eV). The limiter is therefore off by default.

5.5 Anderson acceleration of the Picard iteration (optional)

Each time step is solved by Picard iteration: the coefficients are evaluated from the previous iterate x_k and the linear system gives $g_k = G(x_k)$. On fine meshes the iteration has a slow mode localised on the first few nodes at the magnetic axis, in BV_{1s} (`LQ*8`) of all species, whose contraction factor per sweep tends to 1 as the axis element shrinks (0.70, 0.91, 0.98 at `NRMAX` $= 120$, 240, 480 with $\Delta t = 10^{-4}$); at `NRMAX` $= 480$ it takes about 500 sweeps to reach `EPS` $= 10^{-6}$.

With `NANDER` $\neq 0$ the next iterate is instead

$$x_{k+1} = g_k - \sum_j \gamma_j\,(\Delta x_j + \Delta f_j), \qquad \gamma = \arg\min \Big| f_k - \sum_j \gamma_j \Delta f_j \Big|_2,$$

with $f = g - x$ and Δx_j, Δf_j the differences over the last |`NANDER`| sweeps of the step (Anderson mixing with $\beta = 1$). Each variable is scaled by its profile RMS at the first sweep of the step, as in the convergence test. With `NANDER` < 0 the rows of `LQm1`–`LQm5` do not enter the least-squares fit, because near convergence their increments are rounding (for $\dot{\psi}_t$ the floor of Sec. 5.6); the correction is applied to all variables. The history is restarted at every time step, and dropped (the plain g_k is used) if the least-squares solve fails or returns $|\gamma| > 10^3$. Once the adaptive relaxation of the step has found the iteration not contracting, the rest of the step is left to the relaxation alone: the linear model behind the mixing does not hold there, and with the mixing kept on the first steps from the initial profiles at `NRMAX` $= 960$ end in a negative density. A fixed point of G is a fixed point of

the accelerated iteration, so the converged solution is that of plain Picard; NANDER = 0 (default) is bit-identical to the plain iteration.

With NANDER = −20 (100 steps from $t = 0.02$, $\Delta t = 10^{-4}$, EPS = 10^{-6}, ICMAX = 100, X-point equilibrium) the average number of sweeps falls from 22.7, 36.4, 99.4 to 16.0, 28.4, 58.8 at NRMAX = 60, 120, 240, and the steps that reach ICMAX at NRMAX = 240 from 86 to 9; the circular equilibrium behaves alike. The solution agrees with the plain iteration to within the convergence tolerance. At NRMAX = 480 the LQ*8 mode is removed as well. These figures were measured with the purely relative test of $\dot{\psi}_t$ (LQm2), under which almost every step at NRMAX = 480 still reached ICMAX on the rounding floor of $\dot{\psi}_t$, which the acceleration does not affect. With the floor term of Sec. 5.6 the steps at NRMAX = 240 and 480 end when the other variables have converged: with NANDER = −20 the average number of sweeps is 33.5 (240) and 67.1 (480), and 3 of 100 steps at 480 reach ICMAX, on the other variables.

5.6 Convergence test of $\dot{\psi}_t$ (LQm2)

The sweeps of a time step stop when, for every variable X_q, $\max_j |\Delta X_{q,j}| \le \mathtt{EPS}\,\mathrm{rms}(X_q)$, ΔX_q being the change over the sweep. For $X_2 = \dot{\psi}_t$ this relative test has a floor that no iteration removes, and the test is

$$\max_j |\Delta X_{2,j}| \le \mathtt{EPS}\,\mathrm{rms}(X_2) + C_\mathrm{r}\,\epsilon_\mathrm{m}\,\mathtt{NRMAX}^2\,\frac{\mu_0\,\mathrm{rms}(X_5)}{\mathtt{ADV}\,\Delta t}, \qquad C_\mathrm{r} = 1,$$

with $X_5 = \psi_t/\mu_0$ (LQm5), ϵ_m the machine epsilon and ADV the weight of the implicit terms (2/3 for BDF2, 1 in the first step). LQm2 is not removed from the test: the relative term is kept as it is for every variable, and only the floor term is added. The test of every other variable (including $\dot{\psi}$, LQm3), the equations and the fixed point are unchanged.

The floor has two factors. ψ_t is mostly vacuum flux, and in Ampère's law (the LQm2 row) the $\nabla \times B_\phi$ term applied to it is a cancellation of terms larger than the result, $|KX_5|/(|K||X_5|) = 3.4 \times 10^{-7}$ at NRMAX = 60 falling as $\mathtt{NRMAX}^{-2}$. A rounding of relative size ϵ_m in that row, in the assembly or in the solve, leaves ψ_t determined only to $(0.05\text{–}0.07)\,\epsilon_\mathrm{m}\mathtt{NRMAX}^2$. Faraday's law (the LQm5 row) then gives $\dot{\psi}_t$ as the difference quotient $\mu_0(X_5 - X_5^\mathrm{old})/(\mathtt{ADV}\,\Delta t)$, which multiplies that rounding by $\psi_t/(\mathtt{ADV}\,\Delta t\,|\dot{\psi}_t|)$, 7×10^6 at $\Delta t = 10^{-4}$, $t = 0.02\,\mathrm{s}$, and without bound as ψ_t approaches a steady state. The relative floor of $\dot{\psi}_t$ is therefore about 4×10^{-7}, 1.5×10^{-6}, 4.6×10^{-6}, 1.8×10^{-5} at NRMAX = 60, 120, 240, 480 ($\Delta t = 10^{-4}$, $t = 0.02\,\mathrm{s}$), grows as $1/\Delta t$, and grows in time: at NRMAX = 60 it reaches 10^{-5} by $t = 0.5\,\mathrm{s}$. At the floor the equations are satisfied to rounding (backward error of the LQm2 and LQm5 rows 10^{-15}–10^{-11}, normwise residual of every equation below 4×10^{-9}); the increment of $\dot{\psi}_t$ is the forward error of an ill-conditioned component, not an unconverged iteration. The second term of the test is that floor with a safety factor.

C_r is not a physical tolerance on $\dot{\psi}_t$. $C_\mathrm{r} = 0.3$ lies on the largest floor observed (the rounding of ψ_t reaches 4–6.5 times its geometric mean) and leaves no margin for rounding that depends on the compiler, the BLAS or the order of operations; $C_\mathrm{r} = 1$, three to four times it, is used. The two values give the same numbers of sweeps to within a few per cent and solutions that cannot be told apart. The test on $\dot{\psi}_t$ keeps its role while the iteration converges, since $\dot{\psi}_t$ responds to a change of the currents about 10^4 times more strongly than they change (NRMAX = 60). In the printed and returned convergence measures the value for LQm2 is the part of the relative increment above the floor term.

Against a reference solved with the relative test and up to 300 (100 steps) or 200 (3000 steps) sweeps per step, NANDER = −20, the test with the floor term gives (EPS = 10^{-6}, ICMAX = 100, $\Delta t = 10^{-4}$)

	relative test sweeps / steps at `ICMAX`	with the floor term sweeps / steps at `ICMAX`
X-point, 240, 100 steps	58.8 / 9	33.5 / 0
circular, 240, 100 steps	69.8 / 10	37.1 / 0
X-point, 480, 100 steps	99.6 / 97	67.1 / 3
circular, 480, 100 steps	99.3 / 95	77.0 / 5
X-point, 240, 3000 steps	89.0 / 2167	29.9 / 0
circular, 240, 3000 steps	89.6 / 2093	33.6 / 0

The remaining steps at `ICMAX` are steps on which another variable has not converged. Densities, temperatures, current and Z_{eff} differ from the reference by less than 7×10^{-8}; $\dot{\psi}_t$ and the poloidal electric field derived from it by 0.2–1.5×10^{-4}, within the 0.3–2.1×10^{-4} by which the relative test with `ICMAX` = 100 differs from the same reference, that is, within the floor. The loop voltages and the Poynting-flux terms follow within 6×10^{-7} over 3000 steps, except the rate of change of the poloidal electric-field energy, which is built on $\epsilon_0 \dot{\psi}_t^{\,2}$ and carries the floor of $\dot{\psi}_t$ under either test. Volume-averaged densities, stored energy and current agree to 10^{-10} and the toroidal angular momentum to 2×10^{-8}. The BDF2 errors and orders of n, T, j and Z_{eff} are unchanged (`NRMAX` = 60, $\Delta t = 4 \times 10^{-4}$–$2.5 \times 10^{-5}$, `EPS` = 10^{-8}); only the self-convergence error of $\dot{\psi}_t$ between the two finest steps rises from 6–8×10^{-6} to 1.5×10^{-5}, its floor. With `NANDER` = 0 the slow `LQ*8` mode still decides most steps at `NRMAX` $\geq$ 240, and the floor term changes little.

5.7 New defaults and V5.52 compatibility

parameter	V5.52	V5.60	meaning
`FSLP`	0.3	1.0	k_L
`MDLSOL`	(0)	1	ramp width δ
`MDLNZD`	(0)	1	bundle transport with the source-slaved C^0 part
`MDLWBC`	(0)	1	wall density condition
`MDLNOFL`	(0)	0	Larsen limiter on D_{01} (2: element form; 1: legacy nodal form)
`FLlim`	–	0.25	limiter factor α (C^0; D_{01} with `MDLNOFL` = 1, 2)
`RLXNO`	–	0.5	neutral-only relaxation (legacy `MDLNOFL` = 1 only)
`NANDER`	(0)	0	Anderson acceleration of the Picard sweeps

A value in parentheses means that the switch did not exist in V5.52 and that the value reproduces the V5.52 behaviour. These are all the defaults that differ. `parm/txparm.v552default` sets `FSLP` = 0.3, `MDLSOL` = `MDLNZD` = `MDLWBC` = `MDLNOFL` = 0. With it V5.60 is not bit-identical to V5.52.20 (the optimized build schedules the changed routines differently), but one step from the same state gives double-precision states that agree to 10^{-12} (median over the variables) and 10^{-9} at worst, except $\dot{\psi}_t$ at its Picard noise floor (3×10^{-7}); the profile output agrees to its single-precision resolution. The table `data/Carbon-S0F0_20260921.txt` is read at start-up in either case.

5.8 Other changes since V5.51

- `MDLPCK` = 2 (default): block-tridiagonal solver with one step of iterative refinement; the Picard floor falls from about 10^{-5} to 5×10^{-7} on the worst equation.

- `oldmixa` (default 0.5): Picard under-relaxation applied only on time steps whose sweep is measured not to contract.
- BDF2 keeps its history across `CONT`; one backward-Euler step is taken where there is none (cold start, `LOAD`).
- Analytic axis limit of the Pfirsch–Schlüter Fourier factors, $f_m(\cdot, 0) = 0$; $\hat{q}^2$ at the axis from the interior instead of $q(0)^2$.
- $D_{0\mathrm{z}}$ uses the impurity neutral speed V_{0z}.
- `FSTPTM` $= 3$ (option): multi-ion composition channel for the turbulent particle flux.
- Finite-collisionality closure for the higher-order flows; triangularity of grid-traced equilibria regularized; G-EQDSK p' normalization corrected; `ipbtdir` $= 1$ by default.
- Diagnostic switches, off by default: `iaxproj` (near-axis regularity projection), `Dnfac` (particle diffusion in the impurity continuity equation).